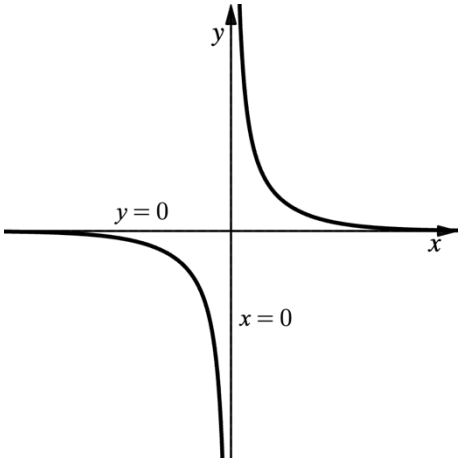
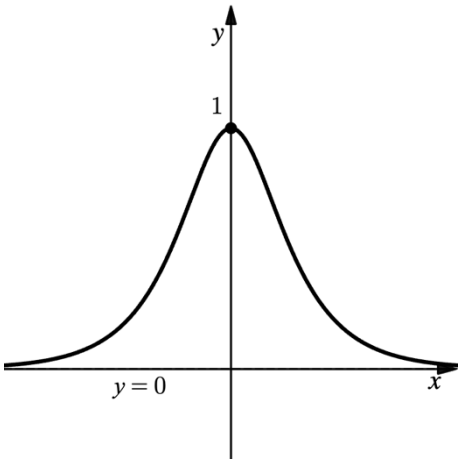
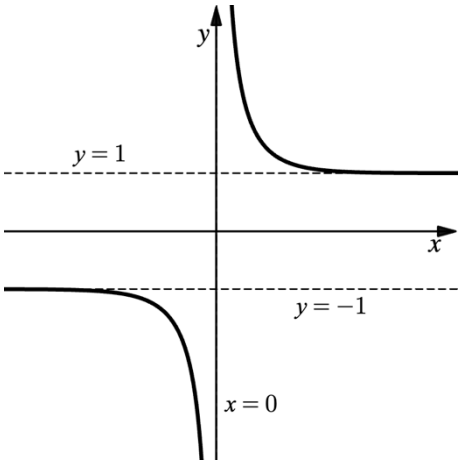
**1) a)**



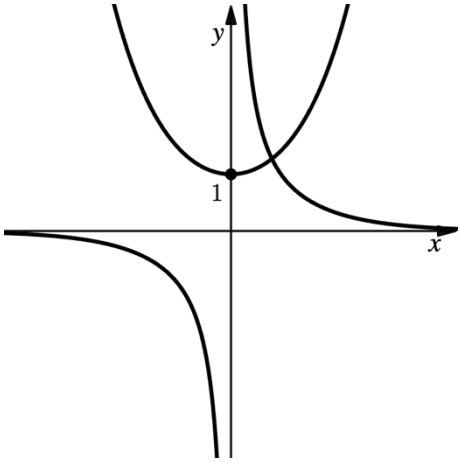

**1) b)**




**1) c)**




**2) a)**




**2) b)**

$\cosh x = \operatorname{cosech} x$

$\Rightarrow \cosh x = \frac{1}{\sinh x}$

$\Rightarrow \sinh x \cosh x = 1$

$\Rightarrow \frac{1}{2}\sinh 2x = 1$

$\Rightarrow \sinh 2x = 2$

$\Rightarrow 2x = \ln\left(2 + \sqrt{2^2 + 1}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(2 + \sqrt{5}\right)$

**Alternatively:**

$\cosh x = \operatorname{cosech} x$

$\Rightarrow \cosh x = \frac{1}{\sinh x}$

$\Rightarrow \sinh x \cosh x = 1$

$\Rightarrow \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2} = 1$

$\Rightarrow (e^x - e^{-x})(e^x + e^{-x}) = 4$

$\Rightarrow e^{2x} - e^{-2x} = 4$

$\Rightarrow e^{2x} - e^{-2x} - 4 = 0$

$\Rightarrow e^{4x} - 1 - 4e^{2x} = 0$

$\Rightarrow e^{4x} - 4e^{2x} - 1 = 0$

$\Rightarrow (e^{2x} - 2)^2 - 4 - 1 = 0$

$\Rightarrow (e^{2x} - 2)^2 = 5$

$\Rightarrow e^{2x} - 2 = \sqrt{5}$

Note that $e^{2x} - 2 \neq -\sqrt{5}$ as $e^{2x} - 2 > -2 > -\sqrt{5}$

$\therefore e^{2x} = 2 + \sqrt{5}$

$\Rightarrow 2x = \ln\left(2 + \sqrt{5}\right)$

$\Rightarrow x = \frac{1}{2}\ln\left(2 + \sqrt{5}\right)$

**3) a)**

The curve must be of the form $y = \operatorname{sech} x + b$

The curve passes through $\left(\ln\left(2+\sqrt{3}\right), 0\right)$:

$0 = \operatorname{sech}\left(\ln\left(2+\sqrt{3}\right)\right) + b$

$\Rightarrow -b = \frac{1}{\cosh\left(\ln\left(2+\sqrt{3}\right)\right)}$

$\Rightarrow \cosh\left(\ln\left(2+\sqrt{3}\right)\right) = -\frac{1}{b}$

$\Rightarrow \frac{e^{\ln\left(2+\sqrt{3}\right)}+e^{-\ln\left(2+\sqrt{3}\right)}}{2} = -\frac{1}{b}$

$\Rightarrow \frac{e^{\ln\left(2+\sqrt{3}\right)}+e^{\ln\left(\left(2+\sqrt{3}\right)^{-1}\right)}}{2} = -\frac{1}{b}$

$\Rightarrow \frac{\left(2+\sqrt{3}\right)+\left(2-\sqrt{3}\right)}{2} = -\frac{1}{b}$

$\Rightarrow 2 = -\frac{1}{b}$

$\Rightarrow b = -\frac{1}{2}$

$\therefore y = \operatorname{sech} x - \frac{1}{2}$

**3) b)**

The curve must be of the form $y = \operatorname{sech}(x-a)$

The curve passes through $\left(0, \frac{2e^3}{1+e^6}\right)$:

$\frac{2e^3}{1+e^6} = \operatorname{sech}(-a)$

Note that $\operatorname{sech}(-a) = \operatorname{sech} a$ as $\operatorname{sech} x$ is an even function

$\frac{2e^3}{1+e^6} = \operatorname{sech} a$

$\Rightarrow \frac{2e^3}{1+e^6} = \frac{1}{\cosh a}$

$\Rightarrow \cosh a = \frac{1+e^6}{2e^3}$

$\Rightarrow a = \pm \ln\left(\frac{1+e^6}{2e^3} + \sqrt{\left(\frac{1+e^6}{2e^3}\right)^2 - 1}\right)$

$\Rightarrow a = \pm \ln\left(\frac{1+e^6}{2e^3} + \sqrt{\frac{e^{12}+2e^6+1}{4e^6} - 1}\right)$

$\Rightarrow a = \pm \ln\left(\frac{1+e^6}{2e^3} + \sqrt{\frac{e^{12}-2e^6+1}{4e^6}}\right)$

$\Rightarrow a = \pm \ln\left(\frac{1+e^6}{2e^3} + \sqrt{\left(\frac{-1+e^6}{2e^3}\right)^2}\right)$

$\Rightarrow a = \pm \ln\left(\frac{1+e^6}{2e^3} + \frac{-1+e^6}{2e^3}\right)$

$\Rightarrow a = \pm \ln\left(\frac{2e^6}{2e^3}\right)$

$\Rightarrow a = \pm \ln(e^3)$

$\Rightarrow a = \pm 3$

The curve has clearly been translated to the left, so $a = -3$

$\therefore y = \operatorname{sech}(x+3)$

**4) a)**

$x = -1,\ y = 2,\ y = 0$

**4) b)**

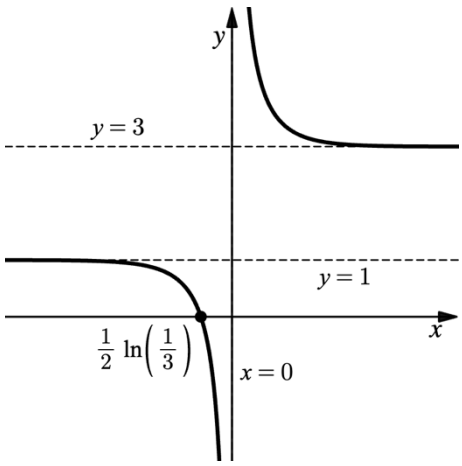
$x = 2,\ y = -2,\ y = -4$

**5) a)**

When $y = 0$:

$0 = \coth x + 2 \Rightarrow \coth x = -2 \Rightarrow \tanh x = -\frac{1}{2}$

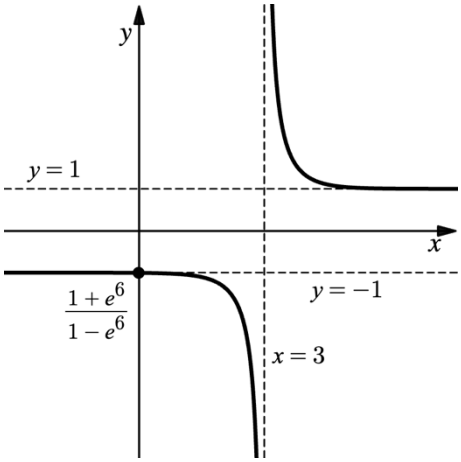
$x = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right) = \frac{1}{2}\ln\frac{1}{3}$




**5) b)**

When $x = 0$:

$y = \coth(-3) = \frac{\cosh(-3)}{\sinh(-3)} = \frac{\frac{e^{-3}+e^3}{2}}{\frac{e^{-3}-e^3}{2}} = \frac{e^{-3}+e^3}{e^{-3}-e^3} = \frac{1+e^6}{1-e^6} \approx -1.005$

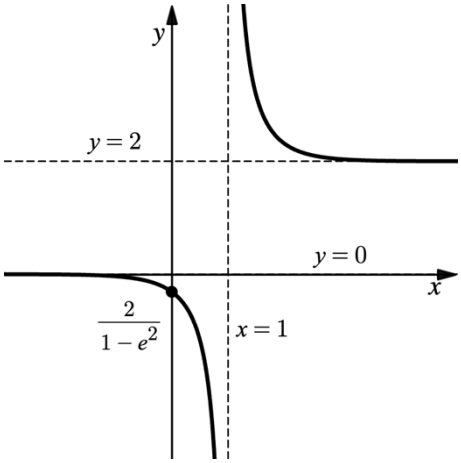



**5) c)**

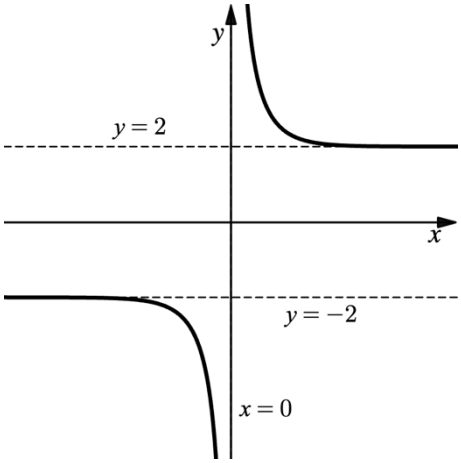
When $x = 0$:

$$y = \coth(-1) + 1 = \frac{\cosh(-1)}{\sinh(-1)} + 1 = \frac{\frac{e^{-1}+e^{1}}{2}}{\frac{e^{-1}-e^{1}}{2}} + 1 = \frac{e^{-1}+e^{1}}{e^{-1}-e^{1}} + 1 = \frac{1+e^{2}}{1-e^{2}} + 1 = \frac{1+e^{2}+1-e^{2}}{1-e^{2}}$$

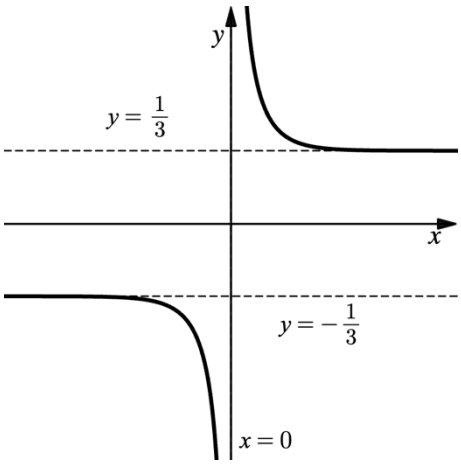
$$= \frac{2}{1-e^{2}} \approx -0.313$$



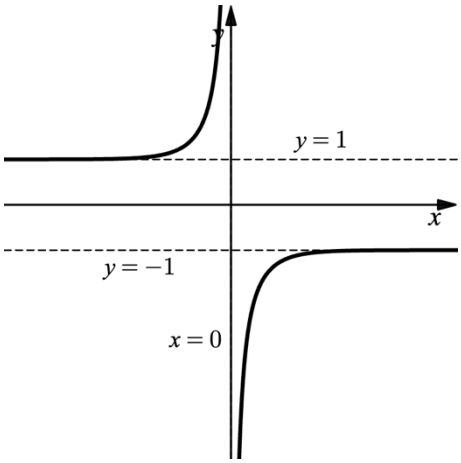

**6) a)**



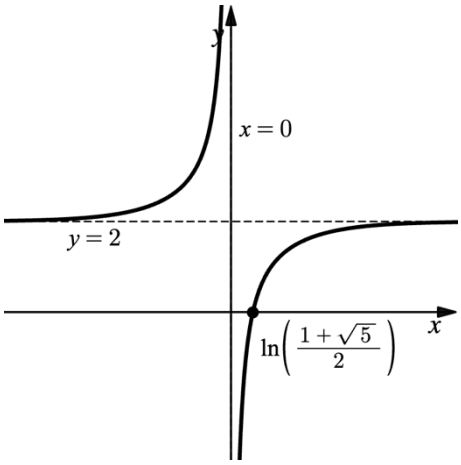

**6) b)**


y
$y = \frac{1}{3}$
x
$y = -\frac{1}{3}$
$x = 0$

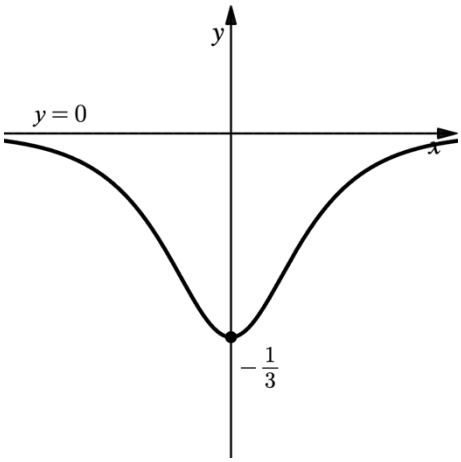
**6) c)**


y
$y = 1$
x
$y = -1$
$x = 0$

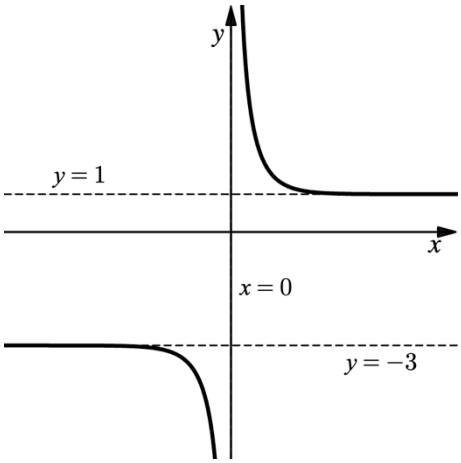
**7) a)**


y
$x = 0$
$y = 2$
x
$\ln\left(\frac{1+\sqrt{5}}{2}\right)$

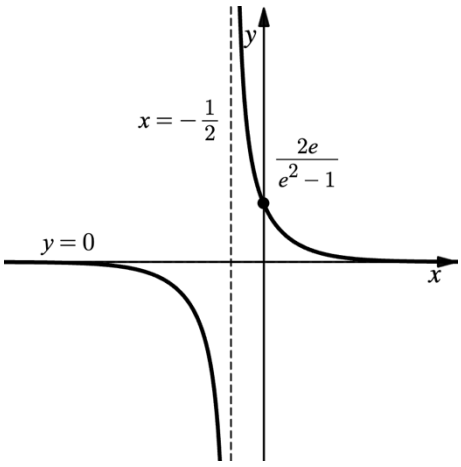
**7) b)**



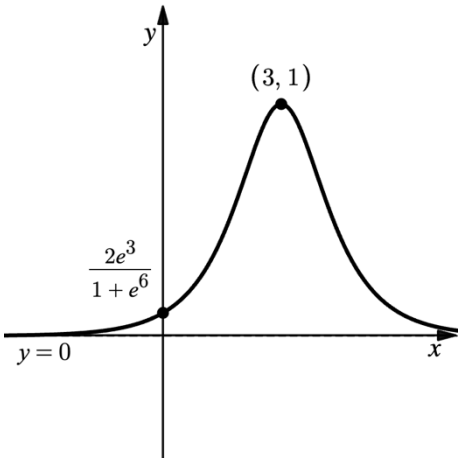

**7) c)**



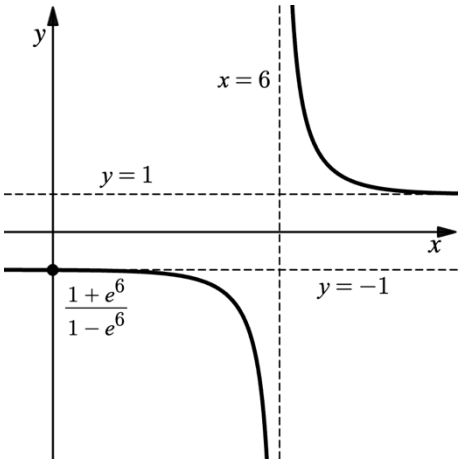

**7) d)**




**7) e)**




**7) f)**




**8)**

It appears to be a transformation of the curve $y = \coth x$, including a reflection in either the $x$ or $y$ axes. We know that $\coth x$ is an odd function, where $\coth(-x) = -\coth x$, so we can focus our attention on a reflection in the $y$-axis without causing problems.

The curve has horizontal asymptotes at $y = 1$ and $y = -1$, so no vertical stretches or translations have been applied.

The curve has a vertical asymptote at $x = 8$, which might suggest $y = \coth(-x - 8)$. However, there may have been a horizontal stretch applied, so we must consider $y = \coth(ax + b)$, where we will expect $a$ to be a negative constant.

Because of the vertical asymptote at $x = 8$, we must have that $8a + b = 0$

The curve passes through $\left(0, \frac{e^8+1}{e^8-1}\right)$ so:

$\frac{e^8+1}{e^8-1} = \coth b$

$\Rightarrow \frac{e^8+1}{e^8-1} = \frac{1}{\tanh b}$

$\Rightarrow \tanh b = \frac{e^8-1}{e^8+1}$

$\Rightarrow b = \frac{1}{2}\ln\left(\frac{1+\frac{e^8-1}{e^8+1}}{1-\frac{e^8-1}{e^8+1}}\right)$

$= \frac{1}{2}\ln\left(\frac{(e^8+1)+(e^8-1)}{(e^8+1)-(e^8-1)}\right)$

$= \frac{1}{2}\ln\left(\frac{2e^8}{2}\right)$

$= \frac{1}{2}\ln(e^8)$

$= \frac{1}{2}\times 8$

$= 4$

If $b = 4$ then $8a + 4 = 0 \Rightarrow a = -\frac{1}{2}$

$\therefore y = \coth\left(-\frac{1}{2}x + 4\right)$